



IBM Data Science Capstone Project – Space X

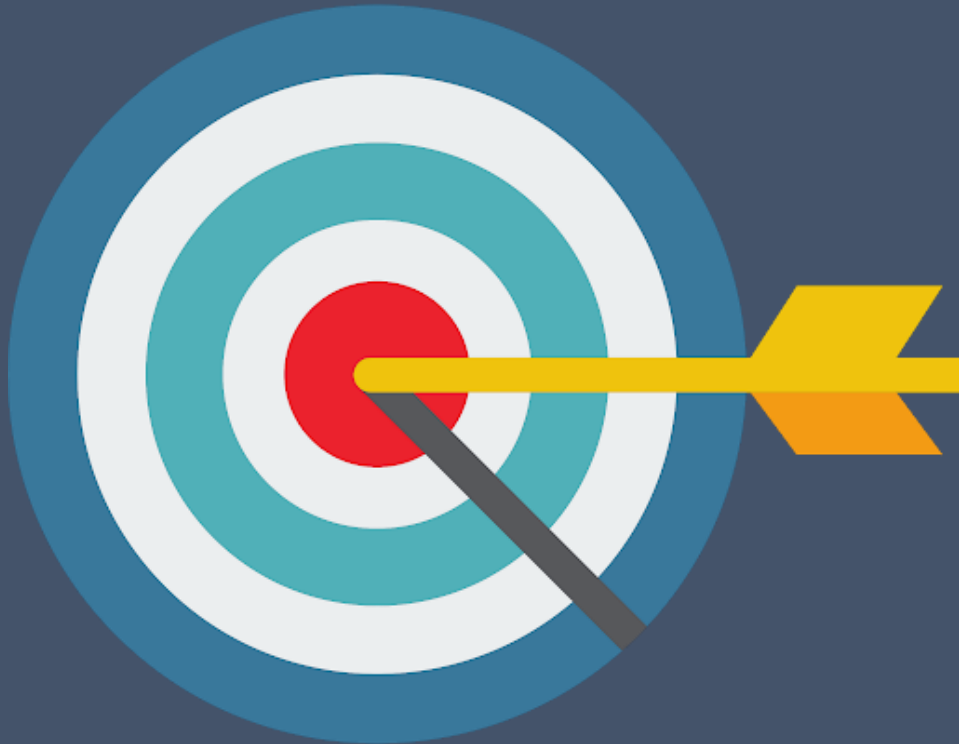
<https://github.com/frbag01/testIBM/blob/main/Data%20Science%20Capstone%20project%20rapporto.pdf>

Outline



- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary



- **Overview del Progetto:** L'obiettivo del progetto è prevedere il successo dell'atterraggio del primo stadio del razzo Falcon 9 di SpaceX per stimare accuratamente i costi di lancio e supportare decisioni commerciali competitive.
 - Data collection
 - Data wrangling
 - EDA with data visualization
 - EDA with SQL
 - Building an interactive map with Folium
 - Building a Dashboard with Plotly Dash
 - Predictive analysis (Classification)
- **Summary of all results** - Exploratory data analysis results - Interactive analytics demo in screenshots - Predictive analysis results
 - **GitHub Repository:**
<https://github.com/frbag01/testIBM>

Introduction



SpaceX garantisce lanci nello spazio a un costo fisso di circa 62 milioni di dollari, rispetto ai 165 milioni di altri fornitori. Gran parte di questo risparmio deriva dal riutilizzo del primo stadio del razzo Falcon 9

Determinare se il primo stadio del razzo atterrerà con successo consente di calcolare con precisione il costo reale della missione. Questa informazione ha un valore strategico fondamentale per un'azienda concorrente che intenda formulare un'offerta commerciale competitiva contro SpaceX per l'assegnazione di gare d'appalto aerospaziali.

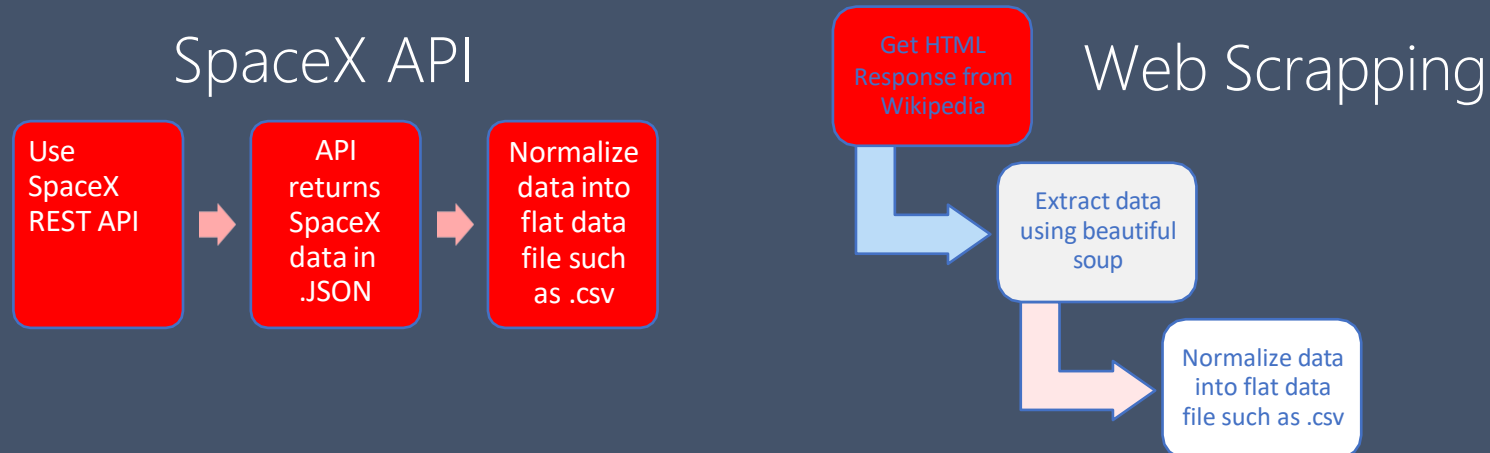
Obiettivo: Costruire una pipeline di Data Science completa (dall'estrazione dati alla modellazione predittiva) per classificare con precisione l'esito dell'atterraggio (Class = 1 successo, Class = 0 fallimento).

Methodology

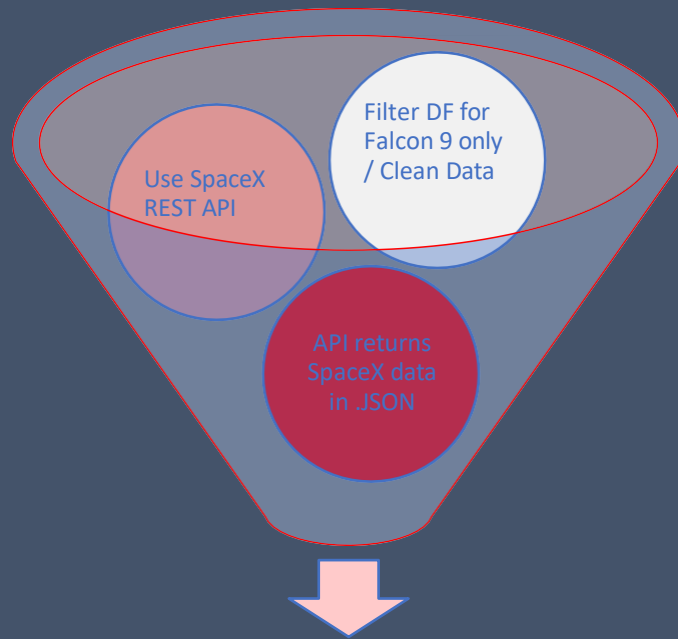
- Data collection methodology:
 - SpaceX Rest API
 - (Web Scrapping) from [Wikipedia](#)
- Performed data wrangling (Transforming data for Machine Learning)
 - One Hot Encoding data fields for Machine Learning and dropping irrelevant columns
- Performed exploratory data analysis (EDA) using visualization and SQL
 - Plotting : Scatter Graphs, Bar Graphs to show relationships between variables to show patterns of data.
- Performed interactive visual analytics using Folium and Plotly Dash
- Performed predictive analysis using classification models
 - How to build, tune, evaluate classification models

Methodology

- The following datasets was collected by
 - We worked with SpaceX launch data that is gathered from the SpaceX REST API.
 - This API will give us data about launches, including information about the rocket used, payload delivered, launch specifications, landing specifications, and landing outcome.
 - Our goal is to use this data to predict whether SpaceX will attempt to land a rocket or not.
 - The SpaceX REST API endpoints, or URL, starts with `api.spacexdata.com/v4/`.
 - Another popular data source for obtaining Falcon 9 Launch data is web scraping Wikipedia using BeautifulSoup.



Data collection – SpaceX API



Normalize data into flat data file such as .csv

- <https://github.com/frbag01/testIBM/blob/main/jupyter-labs-spacex-data-collection-api.ipynb>

simplified flow chart

1. Getting Response from API

```
spacex_url="https://api.spacexdata.com/v4/launches/past"
response = requests.get(spacex_url).json()
```

2. Converting Response to a .json file

```
response = requests.get(static_json_url).json()
data = pd.json_normalize(response)
```

3. Apply custom functions to clean data

```
getLaunchSite(data)
getPayloadData(data)
getCoreData(data)
```

```
getBoosterVersion(data)
```

4. Assign list to dictionary then dataframe

```
launch_dict = {'FlightNumber': list(data['flight_number']),
               'Date': list(data['date']),
               'BoosterVersion':BoosterVersion,
               'PayloadMass':PayloadMass,
               'Orbit':Orbit,
               'LaunchSite':LaunchSite,
               'Outcome':Outcome,
               'Flights':Flights,
               'GridFins':GridFins,
               'Reused':Reused,
               'Legs':Legs,
               'LandingPad':LandingPad,
               'Block':Block,
               'ReusedCount':ReusedCount,
               'Serial':Serial,
               'Longitude': Longitude,
               'Latitude': Latitude}
```

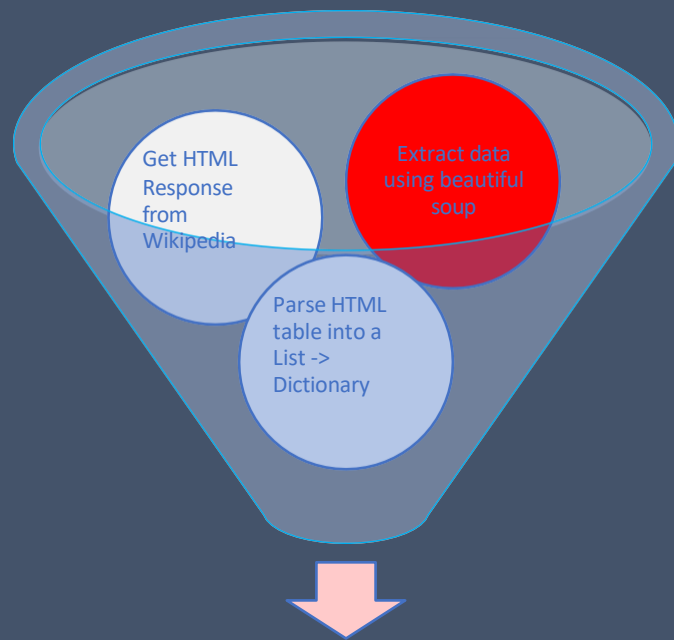
```
df = pd.DataFrame.from_dict(launch_dict)
```

5. Filter dataframe and export to flat file (.csv)

```
data_falcon9 = df.loc[df['BoosterVersion']!="Falcon 1"]
```

```
data_falcon9.to_csv('dataset_part_1.csv', index=False)
```


Data collection – Web Scrapping



Normalize data into flat data file such as .csv

<https://github.com/frbag01/testIBM/blob/main/jupyter-labs-webscraping.ipynb>

1. Getting Response from HTML

```
page = requests.get(static_url)
```

2. Creating BeautifulSoup Object

```
soup = BeautifulSoup(page.text, 'html.parser')
```

3. Finding tables

```
html_tables = soup.find_all('table')
```

4. Getting column names

```
column_names = []
temp = soup.find_all('th')
for x in range(len(temp)):
    try:
        name = extract_column_from_header(temp[x])
        if (name is not None and len(name) > 0):
            column_names.append(name)
    except:
        pass
```

5. Creation of dictionary

```
launch_dict= dict.fromkeys(column_names)

# Remove an irrelevant column
del launch_dict['Date and time ( )']

launch_dict['Flight No.'] = []
launch_dict['Launch site'] = []
launch_dict['Payload'] = []
launch_dict['Payload mass'] = []
launch_dict['Orbit'] = []
launch_dict['Customer'] = []
launch_dict['Launch outcome'] = []
launch_dict['Version Booster']=[[]]
launch_dict['Booster landing']=[[]]
launch_dict['Date']=[[]]
launch_dict['Time']=[[]]
```

6. Appending data to keys (refer) to notebook block 12

```
In [12]: extracted_row = 0
#Extract each table
for table_number,table in enumerate(
    # get table row
    for rows in table.find_all("tr")
    #check to see if first table
```

7. Converting dictionary to dataframe

```
df = pd.DataFrame.from_dict(launch_dict)
```

8. Dataframe to .CSV

```
df.to_csv('spacex_web_scraped.csv', index=False)
```


Data Wrangling

Introduction

In the data set, there are several different cases where the booster did not land successfully. Sometimes a landing was attempted but failed due to an accident; for example, True Ocean means the mission outcome was successfully landed to a specific region of the ocean while False Ocean means the mission outcome was unsuccessfully landed to a specific region of the ocean. True RTLS means the mission outcome was successfully landed to a ground pad False RTLS means the mission outcome was unsuccessfully landed to a ground pad. True ASDS means the mission outcome was successfully landed on a drone ship False ASDS means the mission outcome was unsuccessfully landed on a drone ship.

We mainly convert those outcomes into Training Labels with 1 means the booster successfully landed 0 means it was unsuccessful.

Process

Perform Exploratory Data Analysis EDA on dataset

Calculate the number of launches at each site

Calculate the number and occurrence of each orbit

Calculate the number and occurrence of mission outcome per orbit type

Export dataset as .CSV

Create a landing outcome label from Outcome column

Work out success rate for every landing in dataset

<https://github.com/frbag01/testIBM/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb>

Each launch aims to an dedicated orbit, and here are some common orbit types:

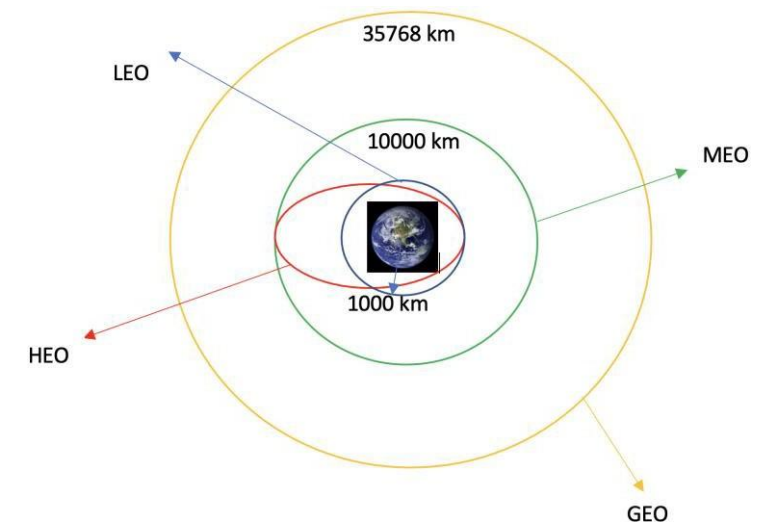


Diagram showing common orbit types SpaceX uses

EDA with Data Visualization

Scatter Graphs being drawn:

Flight Number VS. Payload Mass

Flight Number VS. Launch Site

Payload VS. Launch Site

Orbit VS. Flight Number

Payload VS. Orbit Type

Orbit VS. Payload Mass



Scatter plots show how much one variable is affected by another. The relationship between two variables is called their correlation. Scatter plots usually consist of a large body of data.

Bar Graph being drawn:

Mean VS. Orbit



A bar diagram makes it easy to compare sets of data between different groups at a glance. The graph represents categories on one axis and a discrete value in the other. The goal is to show the relationship between the two axes. Bar charts can also show big changes in data over time.

Line Graph being drawn:

Success Rate VS. Year



Line graphs are useful in that they show data variables and trends very clearly and can help to make predictions about the results of data not yet recorded

<https://github.com/frbag01/testIBM/blob/main/edadataviz.ipynb>

EDA with SQL

Performed SQL queries to gather information about the dataset.

For example of some questions we were asked about the data we needed information about. Which we are using SQL queries to get the answers in the dataset :

- **Displaying the names of the unique launch sites in the space mission**
- **Displaying 5 records where launch sites begin with the string 'KSC'**
- **Displaying the total payload mass carried by boosters launched by NASA (CRS)**
- **Displaying average payload mass carried by booster version F9 v1.1**
- **Listing the date where the successful landing outcome in drone ship was achieved.**
- **Listing the names of the boosters which have success in ground pad and have payload mass greater than 4000 but less than 6000**
- **Listing the total number of successful and failure mission outcomes**
- **Listing the names of the booster_versions which have carried the maximum payload mass.**
- **Listing the records which will display the month names, successful landing_outcomes in ground pad ,booster versions, launch_site for the months in year 2017**
- **Ranking the count of successful landing_outcomes between the date 2010-06-04 and 2017-03-20 in descending order.**



https://github.com/frbag01/testIBM/blob/main/jupyter-labs-eda-sql-coursera_sqlite.ipynb)

Building an interactive map with Folium

To visualize the Launch Data into an interactive map. We took the Latitude and Longitude Coordinates at each launch site and added a *Circle Marker around each launch site with a label of the name of the launch site.*

We assigned the dataframe `launch_outcomes(failures, successes)` to *classes 0 and 1* with **Green and **Red** markers on the map in a `MarkerCluster()`**

Using Haversine's formula we calculated the distance from the Launch Site to various landmarks to find various trends about what is around the Launch Site to measure patterns. **Lines** are drawn on the map to measure distance to landmarks

Example of some trends in which the Launch Site is situated in.

- Are launch sites in close proximity to railways? No
- Are launch sites in close proximity to highways? No
- Are launch sites in close proximity to coastline? Yes
- Do launch sites keep certain distance away from cities? Yes

https://github.com/frbag01/testIBM/blob/main/lab_jupyter_launch_site_location.ipynb

Built an interactive dashboard with Flask and Dash

Used Python Anywhere to host the website live 24/7 so your can play around with the data and view the data

- **The dashboard is built with Flask and Dash web framework.**

Graphs

- **Pie Chart showing the total launches by a certain site/all sites**
 - *display relative proportions of multiple classes of data.*
 - *size of the circle can be made proportional to the total quantity it represents.*

Scatter Graph showing the relationship with Outcome and Payload Mass (Kg) for the different Booster Versions

- It shows the relationship between two variables.
- It is the best method to show you a non-linear pattern.
- The range of data flow, i.e. maximum and minimum value, can be determined.
- Observation and reading are straightforward.

Predictive analysis (Classification)

BUILDING MODEL

- Load our dataset into NumPy and Pandas
- Transform Data
- Split our data into training and test data sets
- Check how many test samples we have
- Decide which type of machine learning algorithms we want to use
- Set our parameters and algorithms to GridSearchCV
- Fit our datasets into the GridSearchCV objects and train our dataset.

EVALUATING MODEL

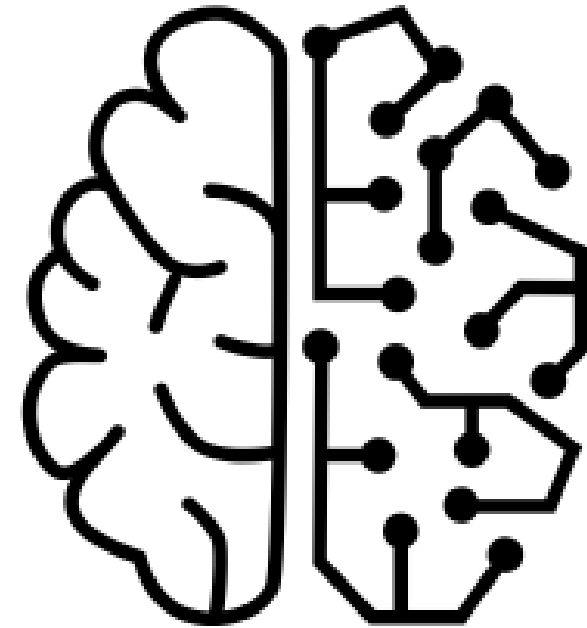
- Check accuracy for each model
- Get tuned hyperparameters for each type of algorithms
- Plot Confusion Matrix

IMPROVING MODEL

- Feature Engineering
- Algorithm Tuning

FINDING THE BEST PERFORMING CLASSIFICATION MODEL

- The model with the best accuracy score wins the best performing model
- In the notebook there is a dictionary of algorithms with scores at the bottom of the notebook.



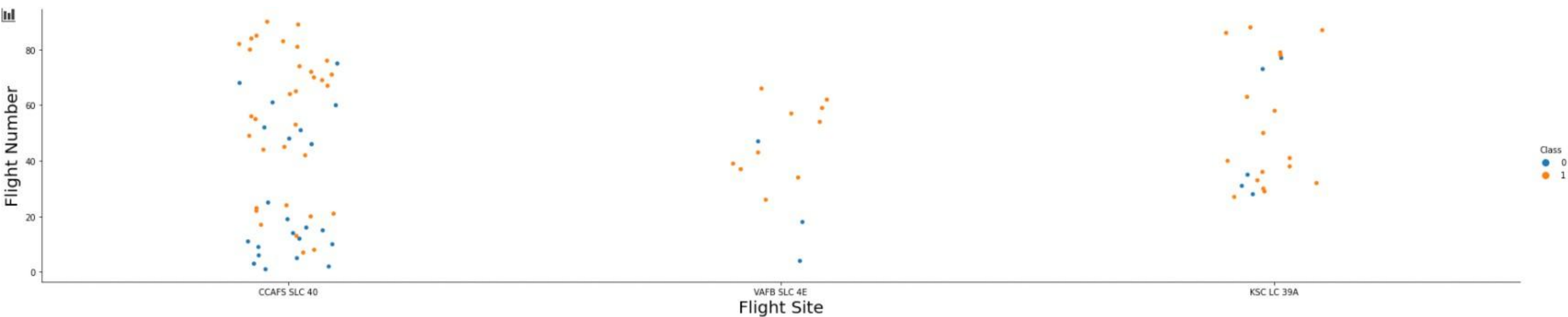
https://github.com/frbag01/testIBM/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

Results



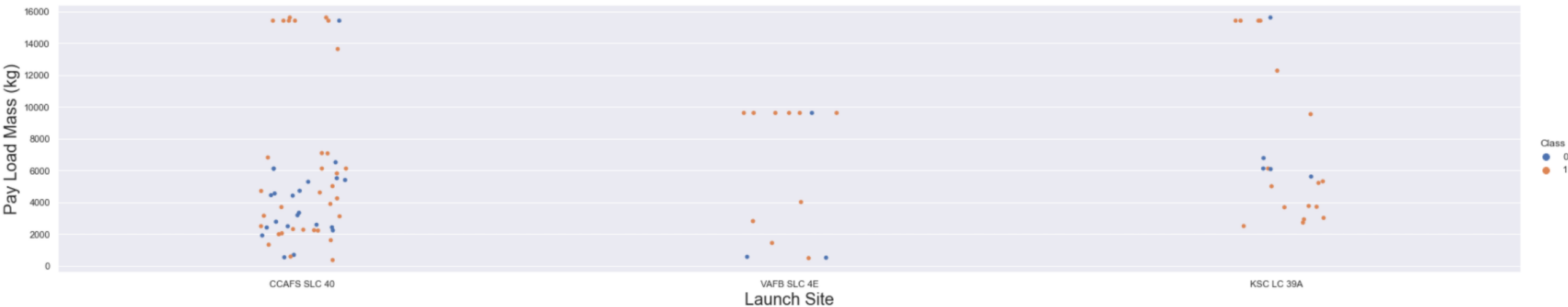
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

Flight Number vs. Flight Site



The more amount of flights at a launch site the greater the success rate at a launch site.

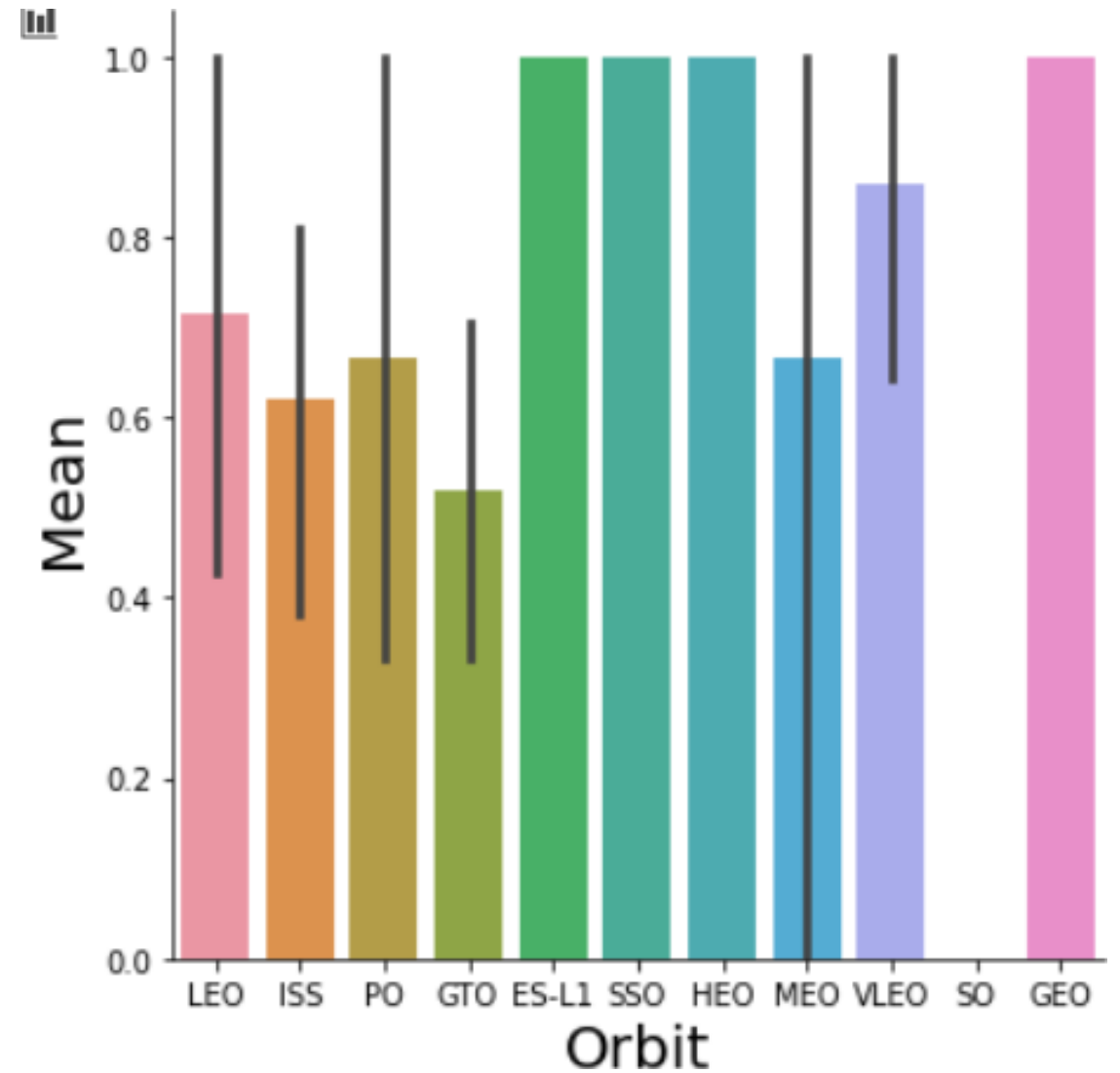
Payload Mass vs. Launch Site



The greater the payload mass for Launch Site CCAFS SLC 40 the higher the success rate for the Rocket. There is not quite a clear pattern to be found using this visualization to make a decision if the Launch Site is dependant on Pay Load Mass for a success launch.

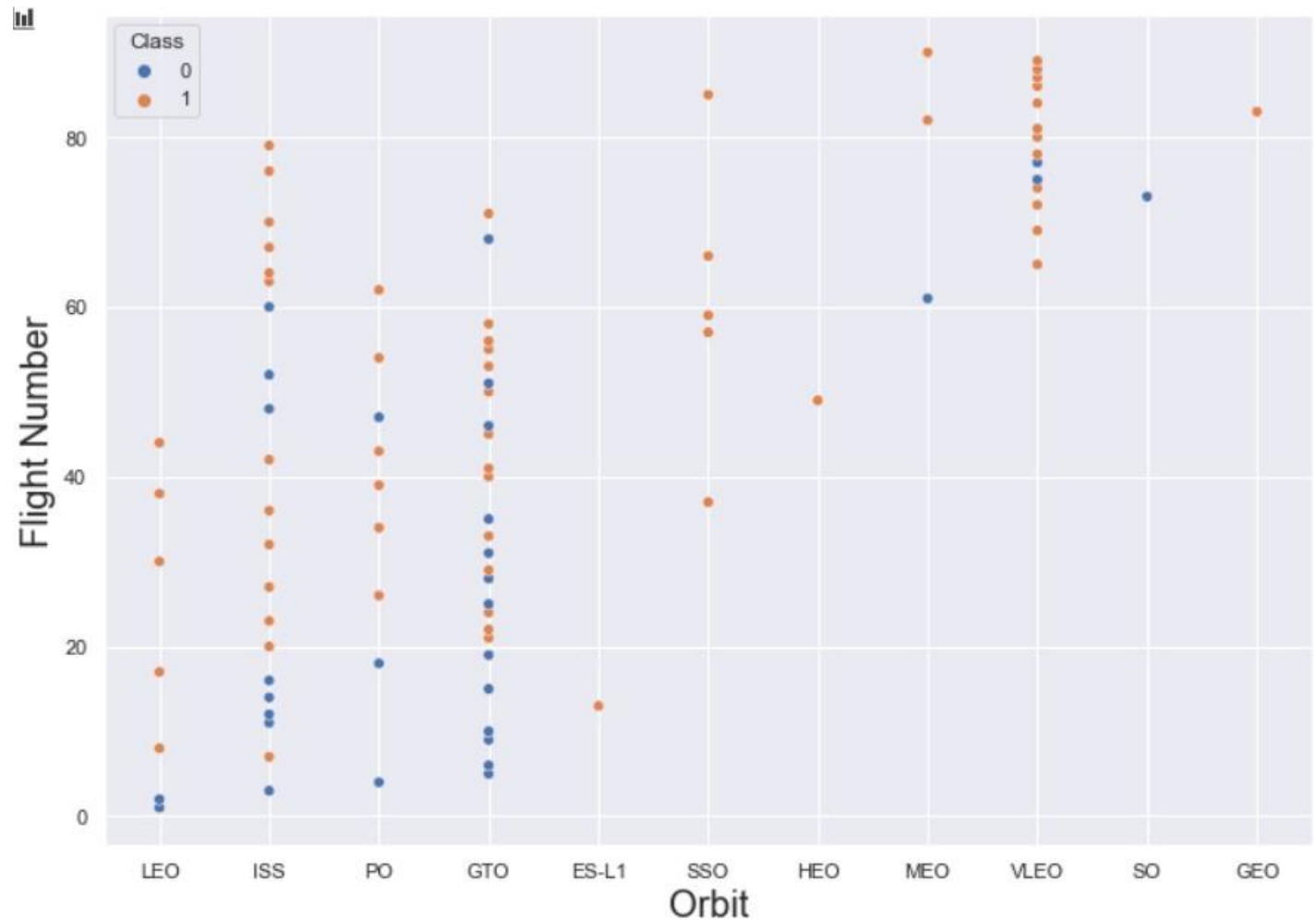
Success rate vs. Orbit type

Orbit GEO,HEO,SSO,ES-L1 has the best Success Rate



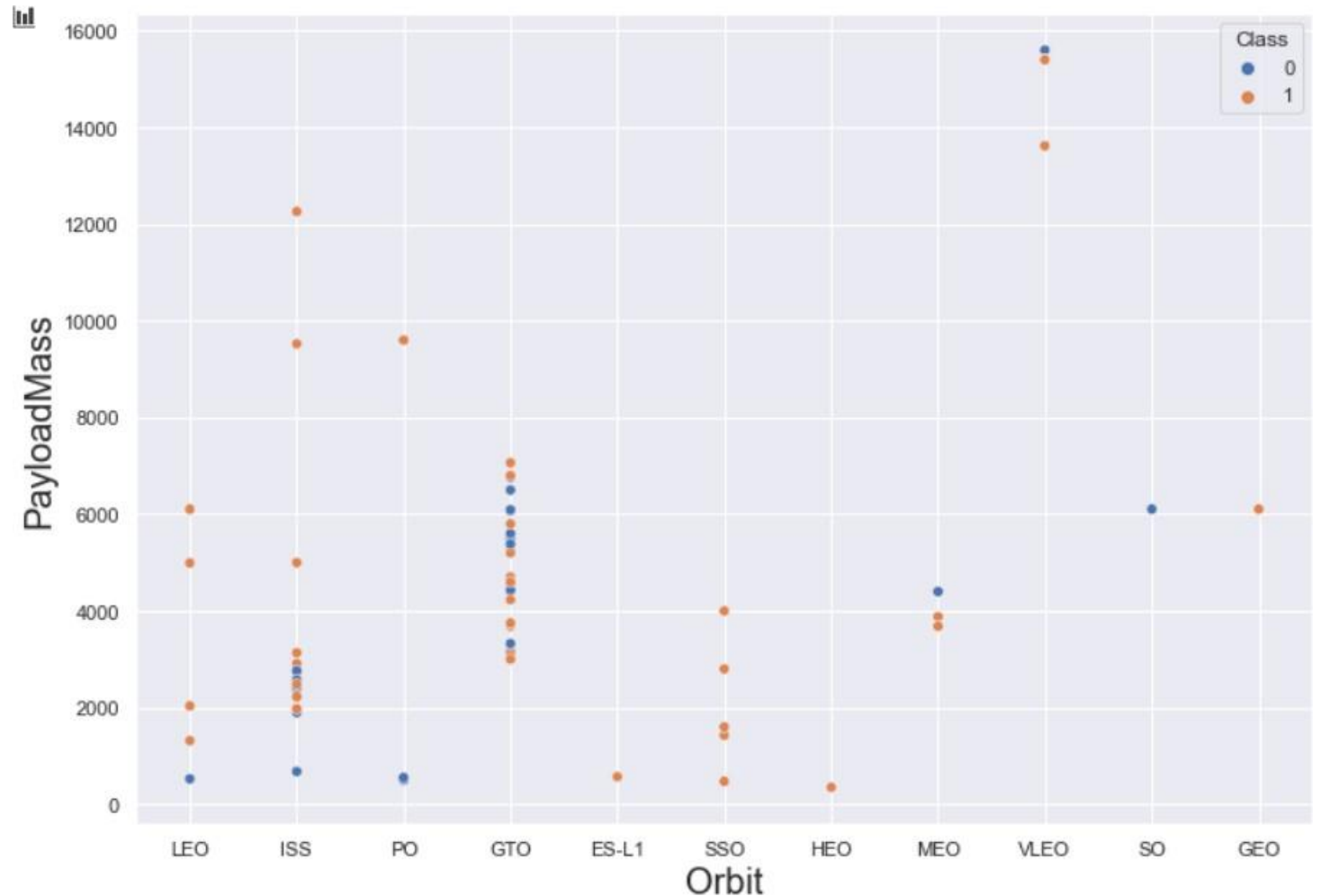
Flight Number vs. Orbit type

You should see that in the LEO orbit the Success appears related to the number of flights; on the other hand, there seems to be no relationship between flight number when in GTO orbit.



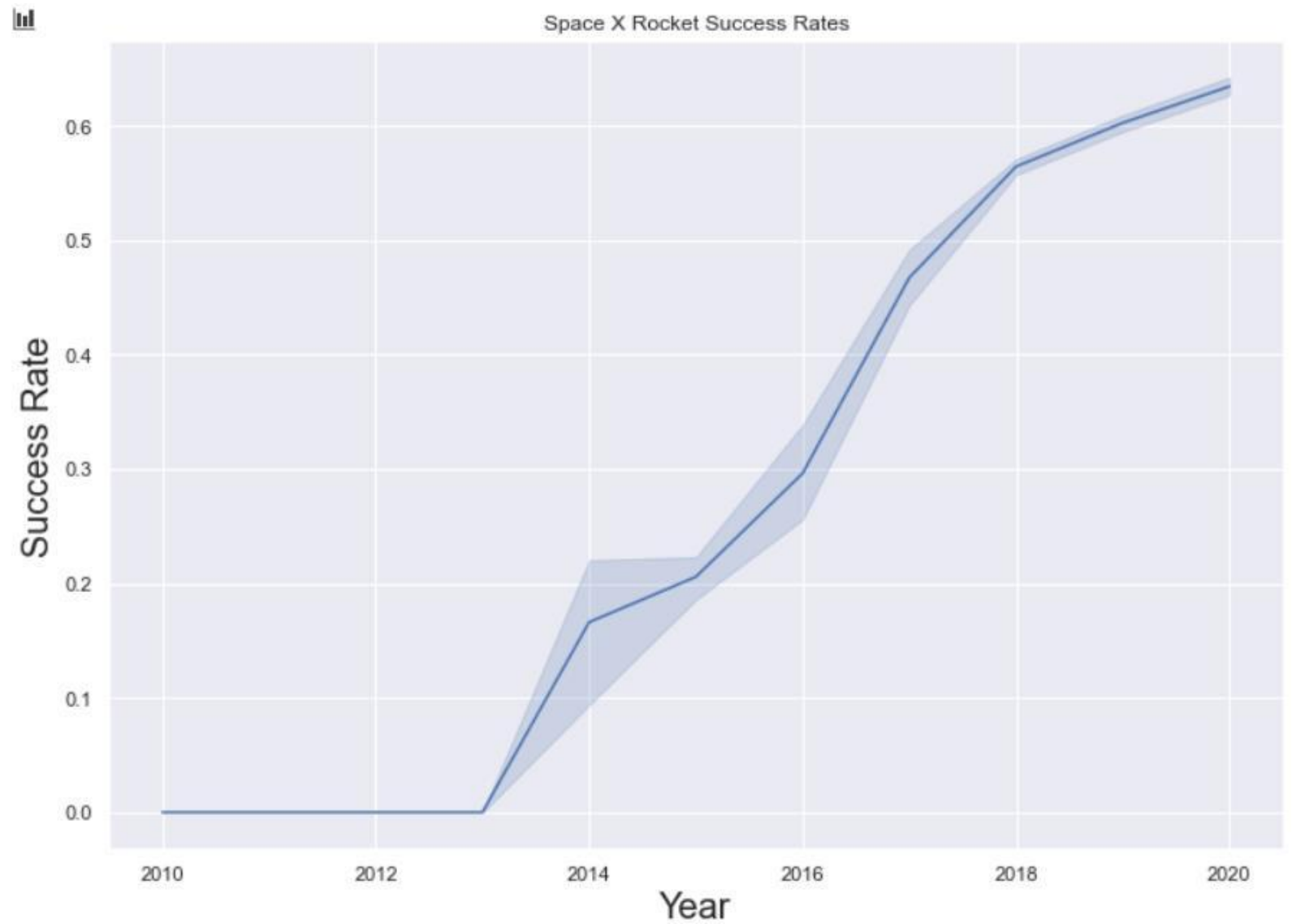
Payload vs. Orbit type

You should observe that Heavy payloads have a negative influence on GTO orbits and positive on GTO and Polar LEO (ISS) orbits.



Launch success yearly trend

you can observe that the success rate since 2013 kept increasing till 2020



Unique Launch Sites

SQL QUERY

```
select DISTINCT Launch_Site  
from tblSpaceX
```



Unique Launch Sites

CCAFS LC-40
CCAFS SLC-40
CCAFS SLC-40
KSC LC-39A
VAFB SLC-4E

QUERY EXPLANATION

Using the word ***DISTINCT*** in the query means that it will only show Unique values in the ***Launch_Site*** column from ***tblSpaceX***

Launch site names begin with `CCA`

SQL QUERY

```
select TOP 5 * from tblSpaceX  
WHERE Launch_Site LIKE 'KSC%'
```



QUERY EXPLANATION

Using the word **TOP 5** in the query means that it will only show 5 records from **tblSpaceX** and **LIKE** keyword has a wild card with the words **'KSC%'** the percentage in the end suggests that the Launch_Site name must start with KSC.

	Date	Time_UTC	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
0	19-02-2017	2021-07-02 14:39:00.0000000	F9 FT B1031.1	KSC LC-39A	SpaceX CRS-10	2490	LEO (ISS)	NASA (CRS)	Success	Success (ground pad)
1	16-03-2017	2021-07-02 06:00:00.0000000	F9 FT B1030	KSC LC-39A	EchoStar 23	5600	GTO	EchoStar	Success	No attempt
2	30-03-2017	2021-07-02 22:27:00.0000000	F9 FT B1021.2	KSC LC-39A	SES-10	5300	GTO	SES	Success	Success (drone ship)
3	01-05-2017	2021-07-02 11:15:00.0000000	F9 FT B1032.1	KSC LC-39A	NROL-76	5300	LEO	NRO	Success	Success (ground pad)
4	15-05-2017	2021-07-02 23:21:00.0000000	F9 FT B1034	KSC LC-39A	Inmarsat-5 F4	6070	GTO	Inmarsat	Success	No attempt

Total Payload Mass by Customer NASA (CRS)

SQL QUERY

```
select SUM(PAYLOAD_MASS_KG_) TotalPayloadMass from tblSpaceX  
where Customer = 'NASA (CRS)';
```

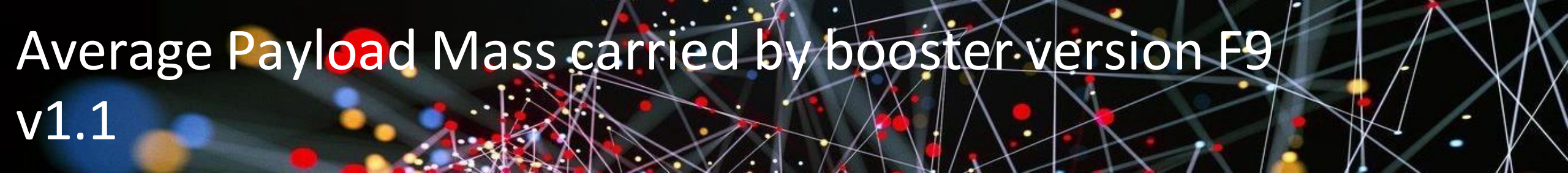


Total Payload Mass	
0	45596

QUERY EXPLANATION

Using the function ***SUM*** summates the total in the column ***PAYLOAD_MASS_KG_***

The ***WHERE*** clause filters the dataset to only perform calculations on ***Customer NASA (CRS)***



Average Payload Mass carried by booster version F9 v1.1

SQL QUERY

```
select AVG(PAYLOAD_MASS_KG_) AveragePayloadMass from tblSpaceX
where Booster_Version = 'F9 v1.1'
```



Average Payload Mass	
0	2928

QUERY EXPLANATION

Using the function **AVG** works out the average in the column **PAYLOAD_MASS_KG_**

The **WHERE** clause filters the dataset to only perform calculations on **Booster_version F9 v1.1**



The date where the successful landing outcome in drone ship was achieved

SQL QUERY

```
select MIN(Date) SLO from tblSpaceX where Landing_Outcome = "Success (drone ship)"
```



Date which first Successful landing outcome in drone ship was acheived.	
0	06-05-2016

QUERY EXPLANATION

Using the function **MIN** works out the minimum date in the column **Date**

The **WHERE** clause filters the dataset to only perform calculations on **Landing_Outcome Success (drone ship)**



Successful drone ship landing with payload between 4000 and 6000

SQL QUERY

```
select Booster_Version from tblSpaceX where Landing_Outcome = 'Success (ground pad)'
AND Payload_MASS_KG_ > 4000 AND Payload_MASS_KG_ < 6000
```



Date which first Successful landing outcome in drone ship was acheived.	
0	F9 FT B1032.1
1	F9 B4 B1040.1
2	F9 B4 B1043.1

QUERY EXPLANATION

Selecting only *Booster_Version*

The **WHERE** clause filters the dataset to *Landing_Outcome = Success (drone ship)*

The **AND** clause specifies additional filter conditions *Payload_MASS_KG_ > 4000 AND Payload_MASS_KG_ < 6000*

Total Number of Successful and Failure Mission Outcomes

SQL QUERY

```
SELECT(SELECT Count(Mission_Outcome) from tblSpaceX where Mission_Outcome  
LIKE '%Success%') as Successful_Mission_Outcomes,  
(SELECT Count(Mission_Outcome) from tblSpaceX where Mission_Outcome  
LIKE '%Failure%') as Failure_Mission_Outcomes
```



QUERY EXPLANATION

a much harder query I must say, we used subqueries here to produce the results. The **LIKE '%foo%'** wildcard shows that in the record the **foo** phrase is in any part of the string in the records for example.

PHRASE "(Drone Ship was a Success)"

LIKE '%Success%'

Word 'Success' is in the phrase the filter will include it in the dataset

Boosters carried maximum payload

SQL QUERY

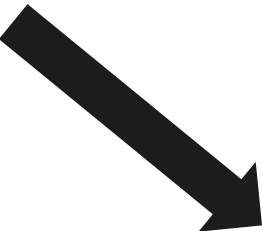
```
SELECT DISTINCT Booster_Version, MAX(PAYLOAD_MASS
_KG_) AS [Maximum Payload Mass]
FROM tblSpaceX GROUP BY Booster_Version
ORDER BY [Maximum Payload Mass] DESC
```

QUERY EXPLANATION

Using the word ***DISTINCT*** in the query means that it will only show Unique values in the ***Booster_Version*** column from ***tblSpaceX***

GROUP BY puts the list in order set to a certain condition.

DESC means its arranging the dataset into descending order



	Booster_Version	Maximum Payload Mass
0	F9 B5 B1048.4	15600
1	F9 B5 B1048.5	15600
2	F9 B5 B1049.4	15600
3	F9 B5 B1049.5	15600
4	F9 B5 B1049.7	15600
...	...	...
92	F9 v1.1 B1003	500
93	F9 FT B1038.1	475
94	F9 B4 B1045.1	362
95	F9 v1.0 B0003	0
96	F9 v1.0 B0004	0
97 rows x 2 columns		

2017 Launch Records

SQL QUERY

```
SELECT DATENAME(month, DATEADD(month,
MONTH(CONVERT(date, Date, 105)), 0) - 1) AS Month,
Booster_Version, Launch_Site, Landing_Outcome
FROM tblSpaceX
WHERE (Landing_Outcome LIKE N'%Success%') AND
(YEAR(CONVERT(date, Date, 105)) = '2017')
```



QUERY EXPLANATION

a much more complex query as I had my **Date** fields in SQL Server stored as **NVARCHAR** the **MONTH** function returns name month. The function **CONVERT** converts **NVARCHAR** to **Date**.

WHERE clause filters **Year** to be 2017

Month	Booster_Version	Launch_Site	Landing_Outcome
January	F9 FT B1029.1	VAFB SLC-4E	Success (drone ship)
February	F9 FT B1031.1	KSC LC-39A	Success (ground pad)
March	F9 FT B1021.2	KSC LC-39A	Success (drone ship)
May	F9 FT B1032.1	KSC LC-39A	Success (ground pad)
June	F9 FT B1035.1	KSC LC-39A	Success (ground pad)
June	F9 FT B1029.2	KSC LC-39A	Success (drone ship)
June	F9 FT B1036.1	VAFB SLC-4E	Success (drone ship)
August	F9 B4 B1039.1	KSC LC-39A	Success (ground pad)
August	F9 FT B1038.1	VAFB SLC-4E	Success (drone ship)
September	F9 B4 B1040.1	KSC LC-39A	Success (ground pad)
October	F9 B4 B1041.1	VAFB SLC-4E	Success (drone ship)
October	F9 FT B1031.2	KSC LC-39A	Success (drone ship)
October	F9 B4 B1042.1	KSC LC-39A	Success (drone ship)
December	F9 FT B1035.2	CCAFS SLC-40	Success (ground pad)



Rank success count between 2010-06-04 and 2017-03-20

SQL QUERY

```
SELECT COUNT(Landing_Outcome)
FROM   tblSpaceX
WHERE  (Landing_Outcome LIKE '%Success%')
AND    (Date > '04-06-2010')
AND    (Date < '20-03-2017')
```



QUERY EXPLANATION

Function **COUNT** counts records in column
WHERE filters data

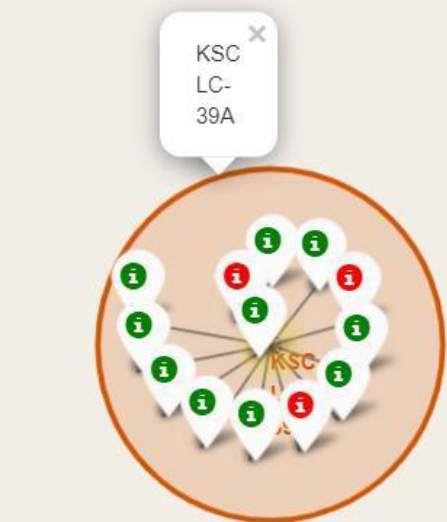
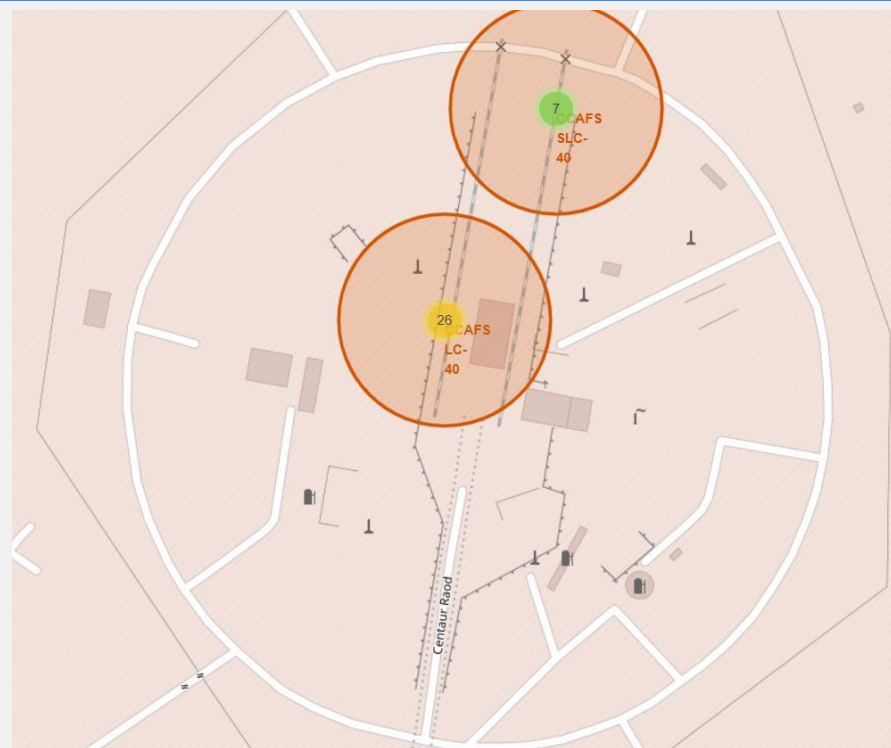
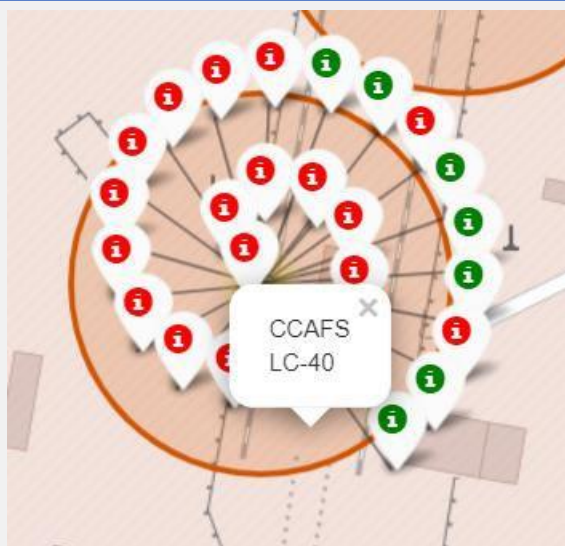
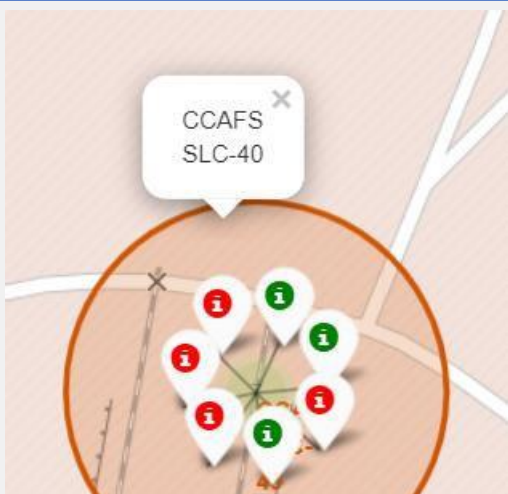
- LIKE** (wildcard)
- AND** (conditions)
- AND** (conditions)

Successful Landing Outcomes Between 2010-06-04 and 2017-03-20	
0	34

All launch sites global map markers



Colour Labelled Markers



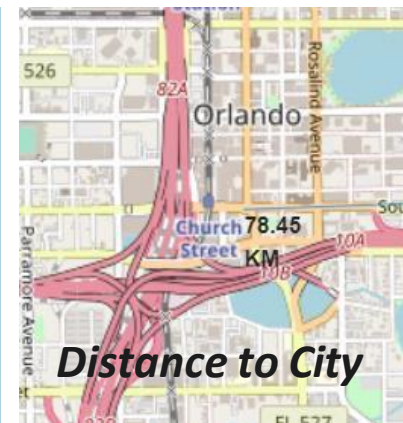
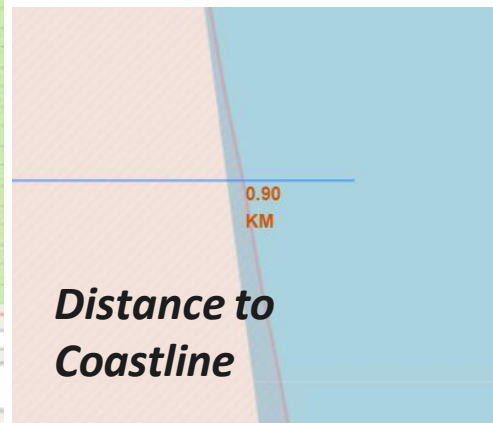
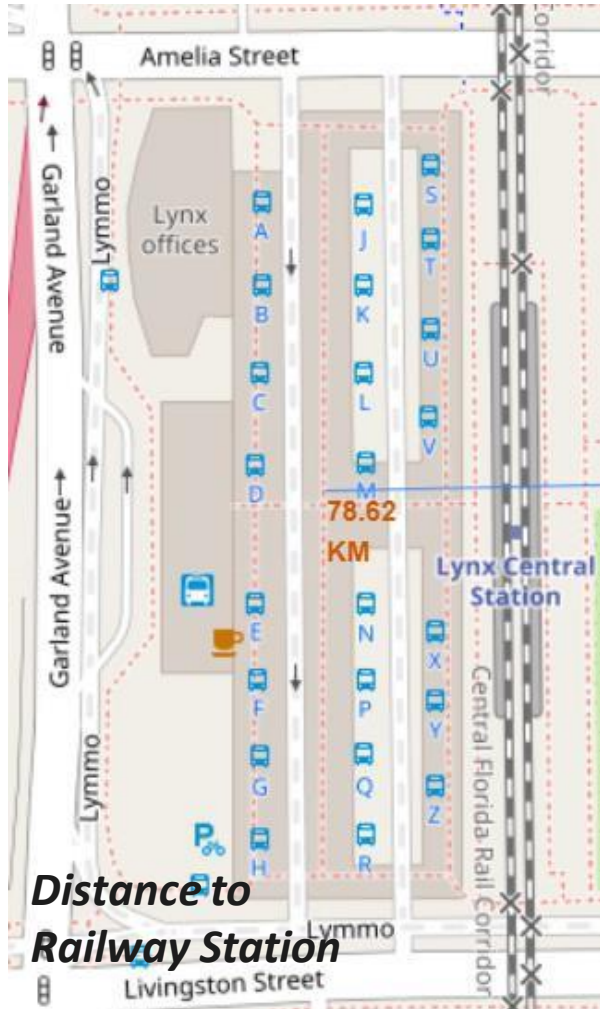
Florida Launch Sites

Green Marker shows successful Launches and *Red Marker* shows Failures



California Launch Site

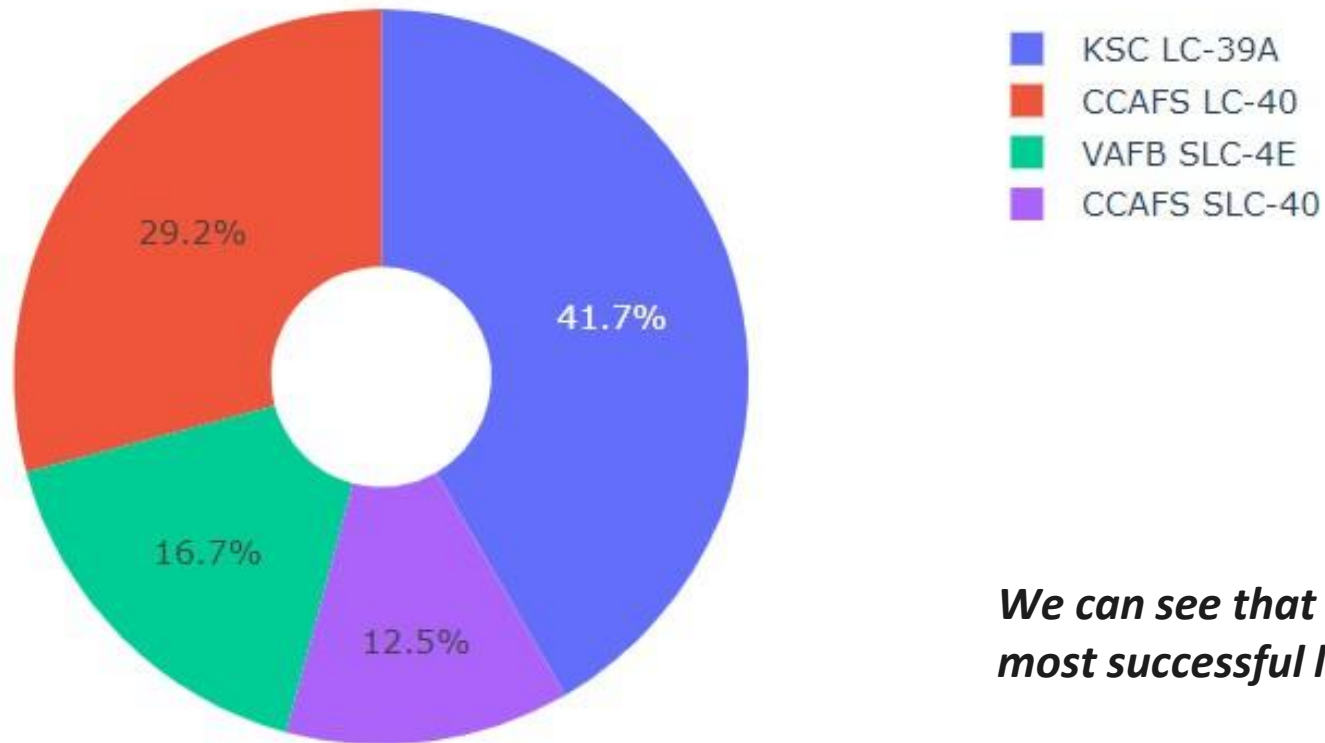
Working out Launch Sites distance to landmarks to find trends with Haversine formula using CCAFS-SLC-40 as a reference



- Are launch sites in close proximity to railways? No
- Are launch sites in close proximity to highways? No
- Are launch sites in close proximity to coastline? Yes
- Do launch sites keep certain distance away from cities? Yes

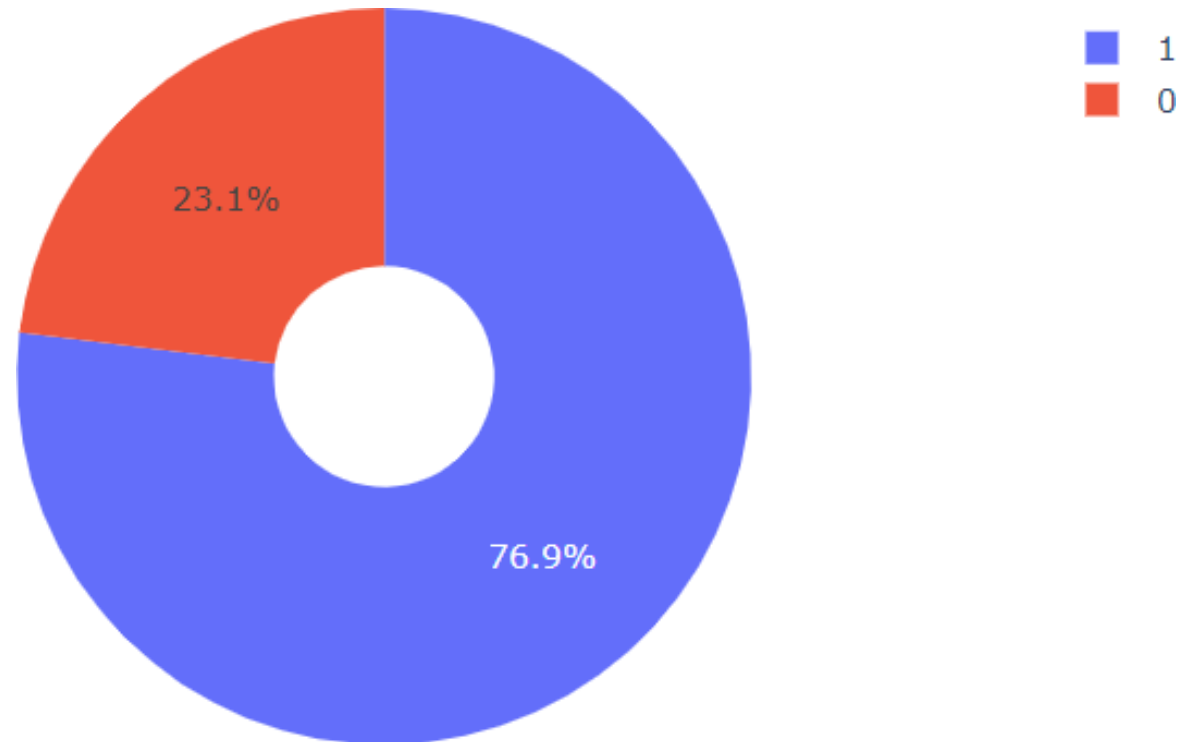
DASHBOARD – Pie chart showing the success percentage achieved by each launch site

Total Success Launches By all sites



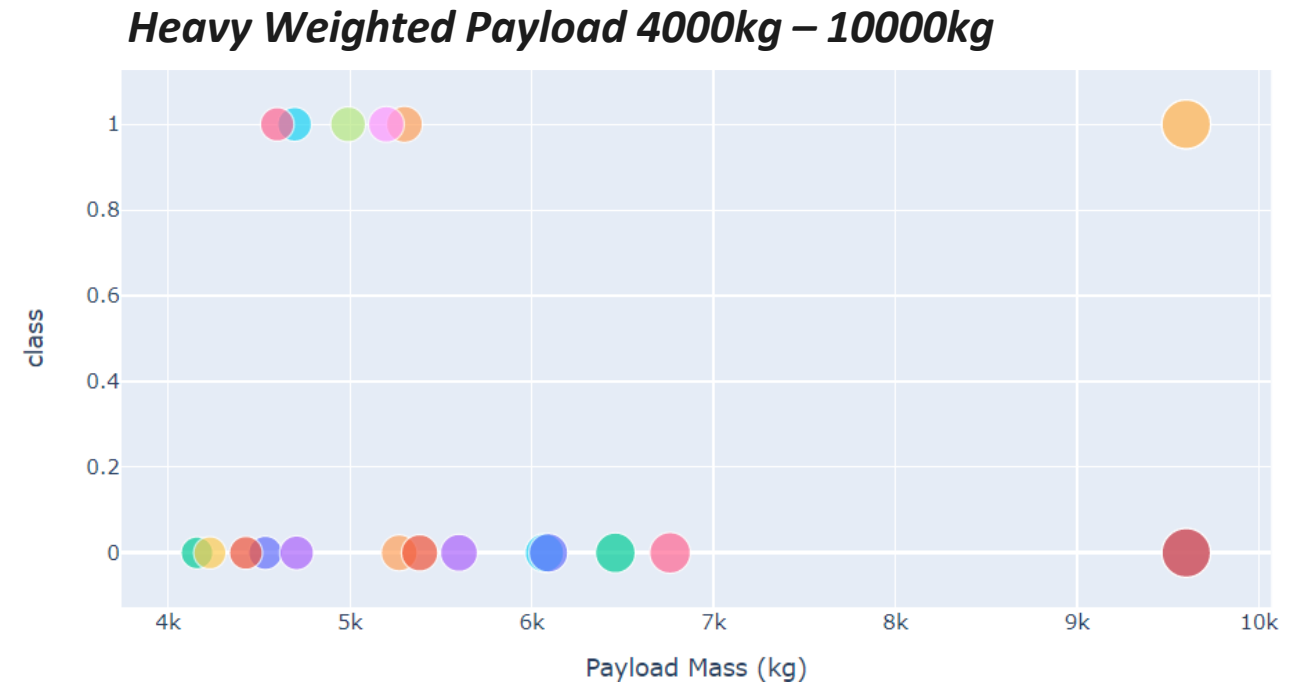
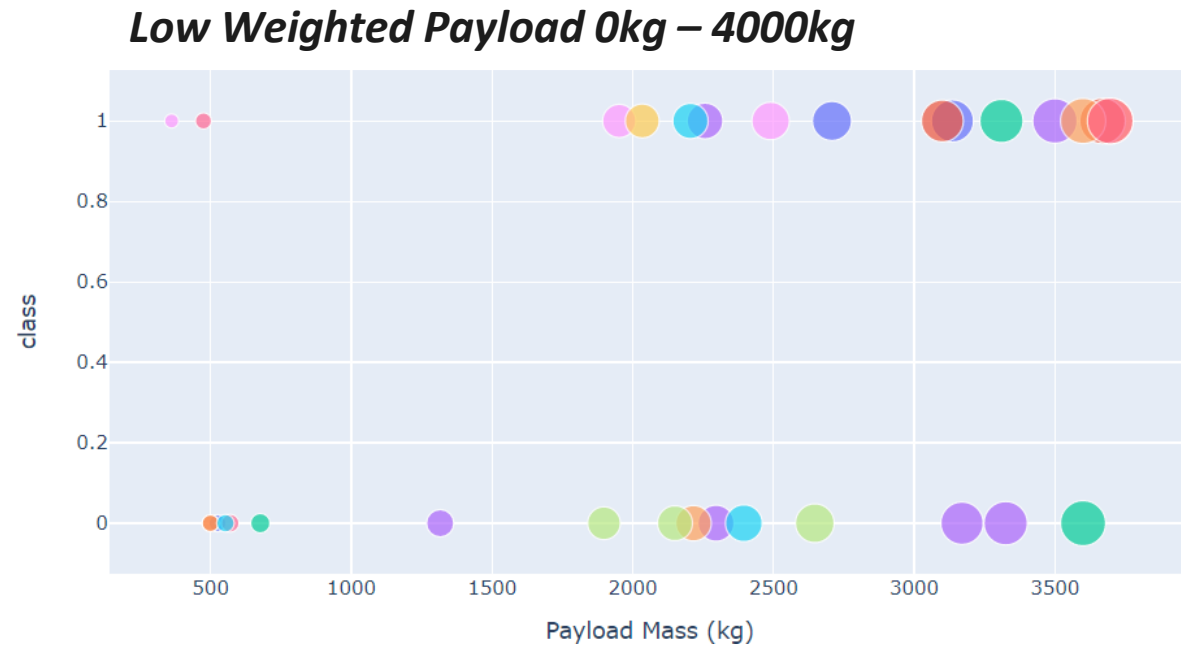
We can see that KSC LC-39A had the most successful launches from all the sites

DASHBOARD – Pie chart for the launch site with highest launch success ratio



KSC LC-39A achieved a 76.9% success rate while getting a 23.1% failure rate

DASHBOARD – Payload vs. Launch Outcome scatter plot for all sites, with different payload selected in the range slider



We can see the success rates for low weighted payloads is higher than the heavy weighted payloads

Classification Accuracy using training data

As you can see our accuracy is extremely close but we do have a winner its down to decimal places! using this function

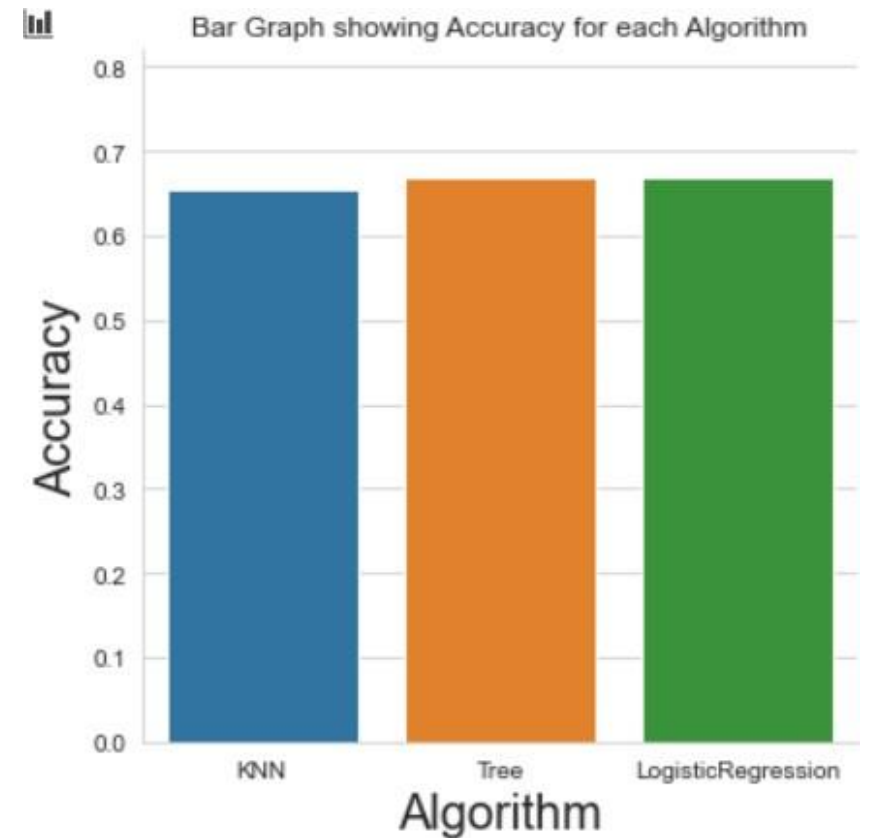
```
bestalgorithm = max(algorithms, key=algorithms.get)
```

	Accuracy	Algorithm
0	0.653571	KNN
1	0.667857	Tree
2	0.667857	LogisticRegression

The tree algorithm wins!!

```
Best Algorithm is Tree with a score of 0.6678571428571429
Best Params is : {'criterion': 'gini', 'max_depth': 2, 'max_features': 'auto', 'min_samples_leaf': 1, 'min_samples_split': 2, 'splitter': 'best'}
```

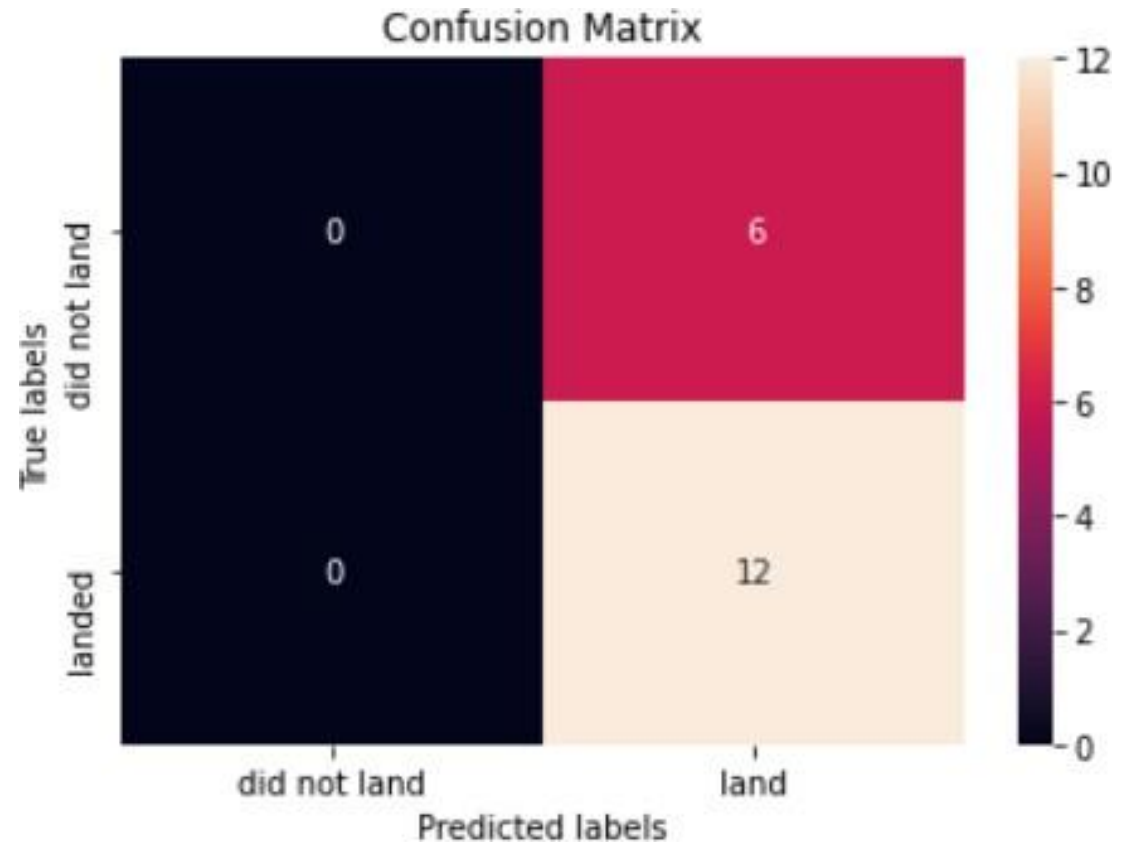
After selecting the best hyperparameters for the decision tree classifier using the validation data, we achieved 83.33% accuracy on the test data.



Confusion Matrix for the Tree

Examining the confusion matrix, we see that Tree can distinguish between the different classes. We see that the major problem is false positives.

		Predicted Values	
		Negative	Positive
Actual Values	Negative	TN	FP
	Positive	FN	TP



Conclusion



- The Tree Classifier Algorithm is the best for Machine Learning for this dataset
- Low weighted payloads perform better than the heavier payloads
- The success rates for SpaceX launches is directly proportional time in years they will eventually perfect the launches
- We can see that KSC LC-39A had the most successful launches from all the sites
- Orbit GEO,HEO,SSO,ES-L1 has the best Success Rate
- <https://github.com/frbag01/testIBM/blob/main/Data%20Science%20Capstone%20project%20rapporto.pdf>